# ADARSH RAJ

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## PROFESSIONAL SUMMARY

B.Tech (AI & ML) student at Birla Institute of Technology, Mesra with hands-on experience building full-stack web applications and backend services. Comfortable across FastAPI/Node.js backends, React frontends, and relational/NoSQL databases, with strong fundamentals in Data Structures & Algorithms, Operating Systems, and DBMS. Seeking Software Engineering internship opportunities.

## EDUCATION

**Birla Institute of Technology, Mesra** **2024 – 2028**
*Bachelor of Technology – Artificial Intelligence and Machine Learning | CGPA: 7.3 / 10*

## PROJECTS

**Business Entity Resolution | GitHub | *Amazon ML Challenge***
- Built an end-to-end entity-resolution pipeline matching ~2.2M Source-1 business records against Source-2/Source-3 records (5M+ each) using only the provided datasets, with no external APIs or lookups.
- Designed blocking over inverted indexes (country, name-token, postal-code rules) to generate candidate pairs and avoid all-pairs comparison; reused the index across train, validation, and test runs.
- Engineered name/address similarity features (Levenshtein, Jaro-Winkler, token Jaccard, TF-IDF cosine) and trained a Logistic Regression classifier with negative sampling; chose the threshold on an entity-level validation split to maximize F0.5.
- Processed data in fixed-size chunks to keep memory bounded at multi-million-record scale; wrote a submission validator checking the required output format.

*Tech Stack: Python, Pandas, Scikit-learn, RapidFuzz, TF-IDF*

**Railway Train Delay Prediction System | GitHub | *Personal Project (In Progress)***
- Building a train-delay prediction system for Indian Railways that combines data science, machine learning, and web development.
- Designed the project architecture and documentation: problem statement, dataset schema, data science pipeline, ML pipeline with evaluation metrics, and API design.
- Planning a FastAPI backend (served with Uvicorn) and a frontend dashboard; planned extensions include On Time / Moderate / Heavy delay classification and live weather and track-congestion data.

*Tech Stack: Python, FastAPI, Uvicorn, Machine Learning, Data Science*

**AI Fraud Prediction | GitHub | *Personal Project***
- Built a fraud-monitoring web application with a FastAPI backend and a React (Vite) frontend for submitting and reviewing transactions in real time.
- Integrated a Scikit-learn Isolation Forest model to flag transactions as potentially fraudulent based on transaction amount, served through a REST API.
- Designed a SQLite persistence layer using SQLAlchemy to store transaction records and prediction outcomes; started from an existing template and customized the backend, frontend, and model integration.

*Tech Stack: Python, FastAPI, SQLAlchemy, SQLite, React, Scikit-learn*

**CoreSched: CPU Scheduling Simulator | GitHub | *Personal Project***
- Extended an open-source C++ CPU scheduling simulator implementing FCFS, Round Robin, and Completely Fair Scheduler (CFS) via the Strategy design pattern.
- Added an output/logging and UI wrapper layer on top of the base simulator to make scheduling results easier to inspect and visualize.

*Tech Stack: C++, CMake, Strategy Design Pattern*

**TENSOR-TITANS: BTC Price Direction Prediction | GitHub | *Hackathon Project (Winner)***
- Won the hackathon as part of a team; certificate of achievement received.
- Collaborated in a team hackathon to build a real-time pipeline predicting short-term BTC price direction (up/down) from streaming price ticks using classical ML (Logistic Regression, Random Forest).
- Built the backend and UI layer of the pipeline, supporting live ingestion, prediction serving, and monitoring views for the team's feature-engineering and modeling components.

*Tech Stack: Python, Scikit-learn, Pandas*

## CERTIFICATIONS

**British Airways Data Science Job Simulation** | Forage | *October 2026*
- Completed practical tasks in modeling lounge eligibility at Heathrow Terminal 3 and predicting customer buying behaviour.

## TECHNICAL SKILLS

**Languages:** Python, JavaScript, C++, SQL
**Backend:** FastAPI, Node.js, Express.js, REST APIs
**Frontend:** React, HTML, CSS, JavaScript
**Databases:** SQLite, MongoDB
**ML / Data:** Machine Learning, Deep Learning, Scikit-learn, Pandas, NumPy
**Developer Tools:** Git, GitHub
**Core CS:** Data Structures & Algorithms, Operating Systems, DBMS